Data Justice & Genomic Surveillance

COLLEGE 117

May 13, 2026

Professor Daphne Martschenko

Professor Nicole Martinez



Guiding Questions:

1. Who benefits and who is harmed by genomic surveillance?

2. What kinds of trade-offs are raised by genomic surveillance?

Genomic Surveillance

- Use of genomics to track infectious diseases by sequencing the genomes of bacteria, parasites and viruses.

- The process of constantly monitoring pathogens and analyzing their genetic similarities and differences.



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Genomic Surveillance

During Covid-19 pandemic, WHO endorsed genomic surveillance as a worldwide priority

Other examples of where genomic surveillance has been useful:

- Ebola virus disease
- influenza



How can GENOMIC SURVEILLANCE help during a pandemic?

Monitoring and analyzing the genetic make-up of pathogens with pandemic and epidemic potential, can help us understand how they behave and evolve.

This allows us to develop appropriate **tools to control the disease**, such as:

- Vaccines
- Therapeutics
- Diagnostics
- Public health and social measures, like masks and handwashing

The Global Genomic Surveillance Strategy
for Pathogens with Pandemic and Epidemic Potential

World Health Organization

Why is GENOMIC SURVEILLANCE important?

Disease detectives and health authorities need different types of data to control outbreaks, including:

- clinical and laboratory data
- epidemiological data
- environmental data
- other contextual data

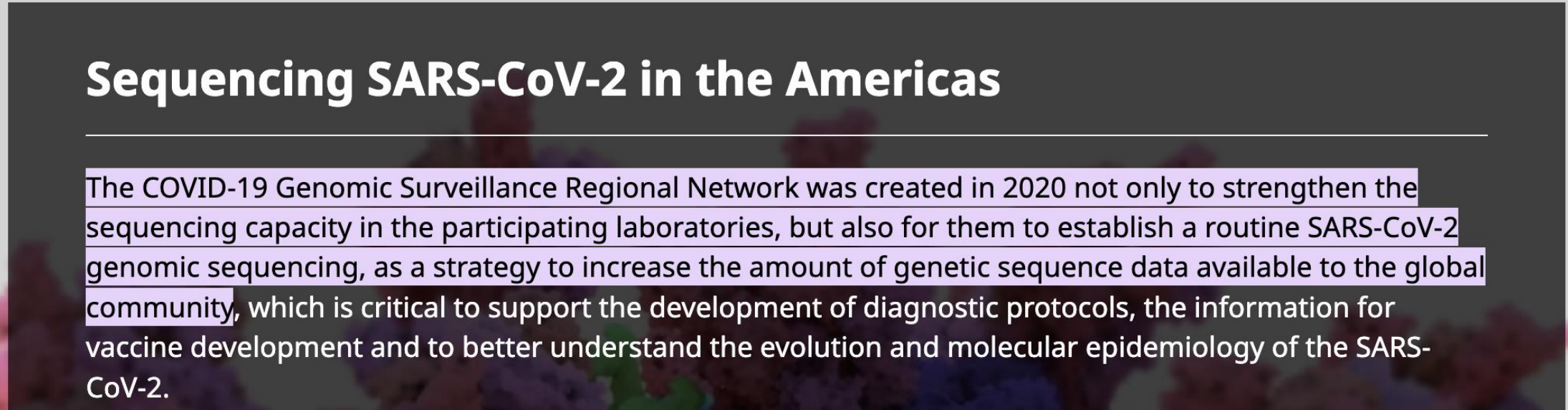
By adding **genomic data**, they can more quickly understand how a pathogen behaves and how to control it.

This is a **powerful tool** in public health surveillance.

The Global Genomic Surveillance Strategy
for Pathogens with Pandemic and Epidemic Potential

World Health Organization

RESVIGEN (Respiratory Virus Genomic Surveillance Regional Network)





AfricaCDC
Centres for Disease Control and Prevention

Our Work | RCCs | Resources | Emergencies | Events

Africa Pathogen Genomics Initiative | Africa PGI 2.0

A flagship initiative of the Africa CDC to build and strengthen a resilient, integrated, proactive, and sustainable molecular diagnostic, genomic surveillance and epidemiology ecosystem in Africa



Global SARS-CoV-2 sequencing capability





Benefits of Genomic Surveillance

1. **Early detection and outbreak tracking**
    - Catching people before they interact with too many others
    - Identifying who has been exposed to an infected person
2. **Vaccine and diagnostic development**
    - Can help identify pathogens to target in vaccines
3. **Tracking antimicrobial resistance (AMR)**
    - Earlier detection of dangerous resistant strains
4. **The more data collected, the more patterns become visible.**
    - Reducing uncertainty increases predictive power.



Benefits of Genomic Surveillance

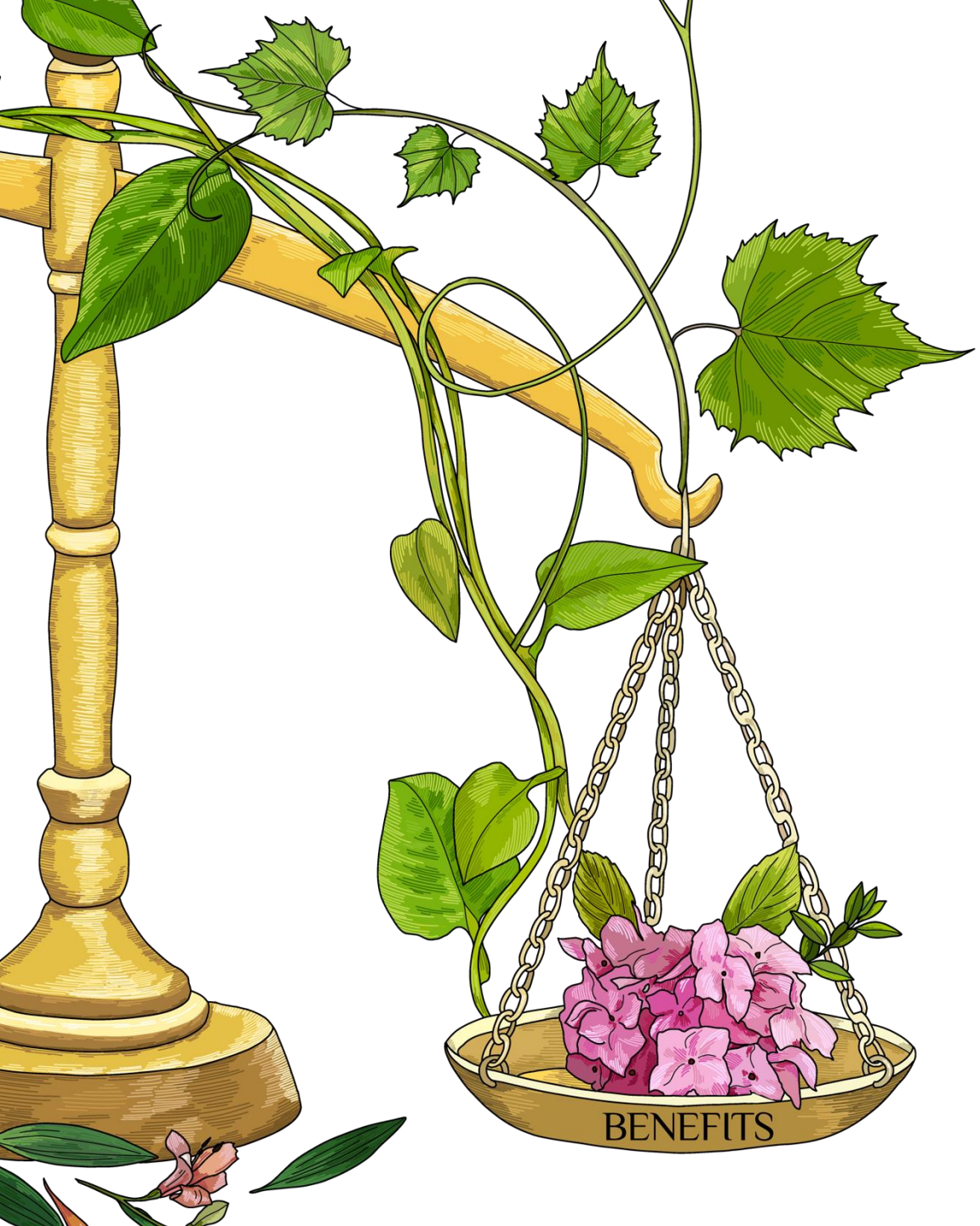
Informing public health policy

- Guide decisions regarding the introduction of pharmacological and non-pharmacological interventions
- Has the promise of overcoming inequalities in access to infectious disease diagnostics and control interventions
- An important tool for early detection of novel and endemic pathogens in low-resource settings (low-resource settings that are disproportionately affected by outbreak risk)

Genomic surveillance as a global public good!




Justice



Justice & Genomic Surveillance

Genomic surveillance remains expensive.

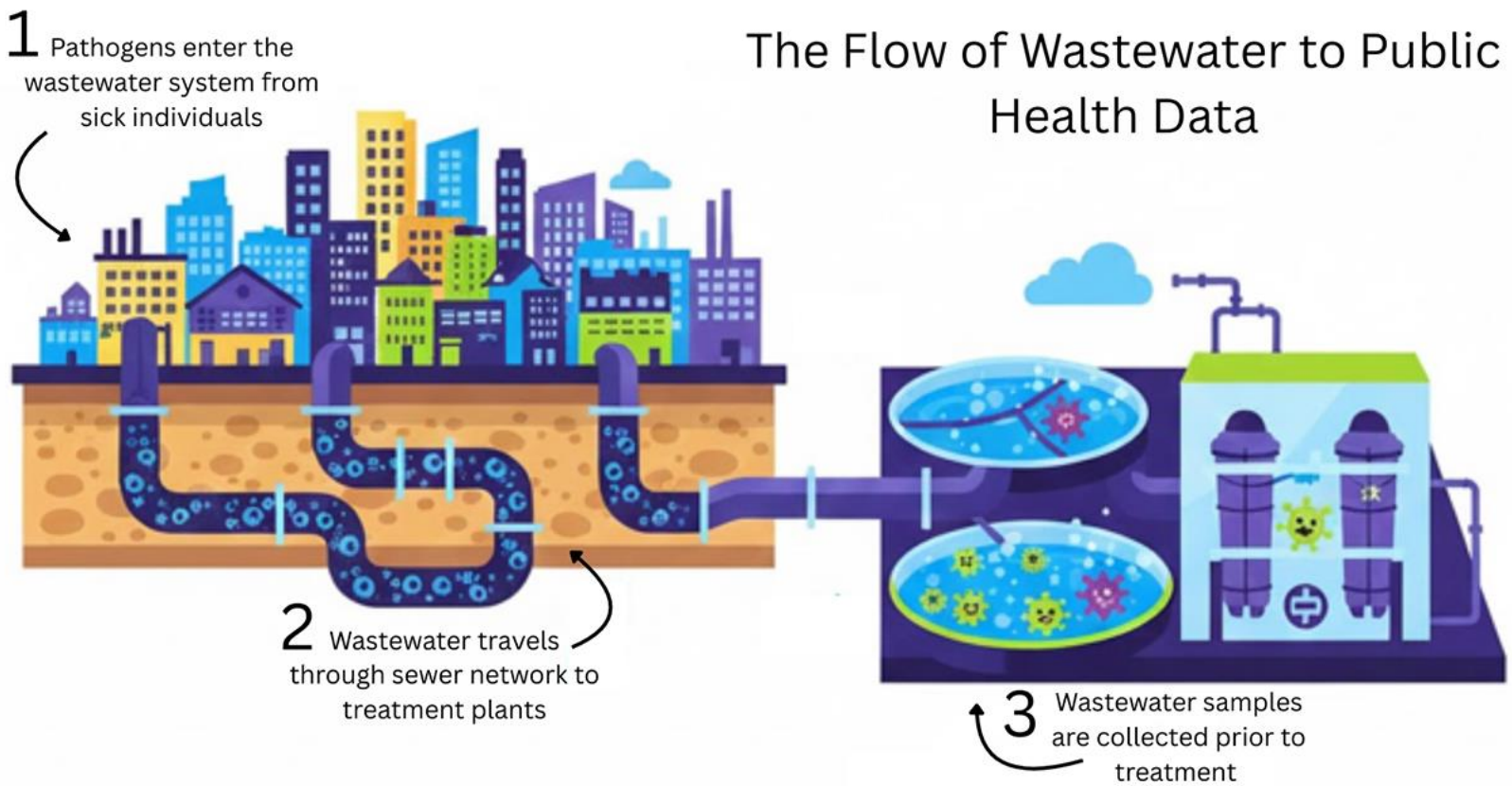
- Initial investments to establish sequencing and bioinformatics capacity are estimated to be US$100,000–$700,000.
- Cost per sequence range from US$20–$200
- **Poor countries can pay up to 10-fold more than high-income countries**

Existing surveillance and laboratory system capacity determines the overall utility of applying genomic surveillance to various pathogens

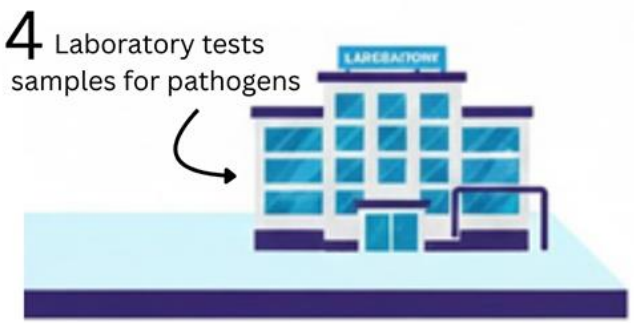
A Surveillance and laboratory system capacity

	LIMITED CAPACITY	MODERATE CAPACITY	HIGH CAPACITY
SURVEILLANCE AND SAMPLING	Passive and ad hoc	Passive and active - moderate coverage	Passive and active - high coverage
LABORATORY AND BIOINFORMATICS	Minimal	Modest scale, slow turnaround time	Well-established, rapid turnaround time
POLICY AND GUIDELINES FOR PATHOGEN GENOMICS	Absent, or present only for SARS-CoV-2	Present for some pathogens	Clearly outlined for endemic and novel pathogens
QUALITY ASSURANCE MECHANISMS	Absent	Some national accreditation; no external quality assurance (EQA)	Well-established national accreditation and EQA
TECHNICAL EXPERT COMMITTEES	Absent	Established	Well-functioning

The Flow of Wastewater to Public Health Data







Covid: South Africa 'punished' for detecting new Omicron variant

27 November 2021

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South Africa has complained it is being punished - instead of applauded - for discovering Omicron, a concerning new variant of Covid-19.

The foreign ministry made the statement as countries around the world restricted travel from southern Africa as details of the spread emerged.

What trade-offs does genomic surveillance raise?

Nobody has responded yet.

Hang tight! Responses are coming in.

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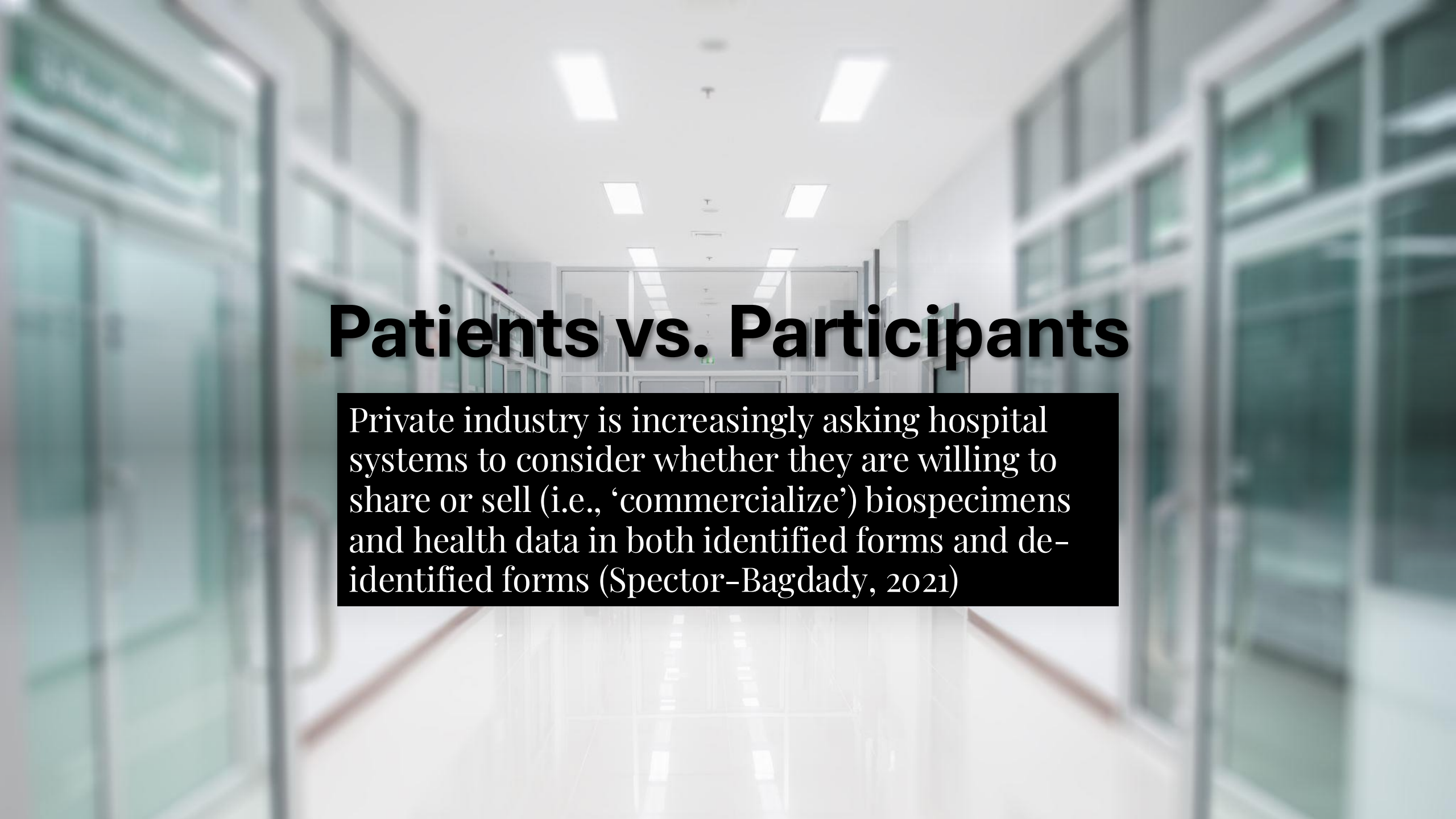


Relationship (or trade-off) between liberty/autonomy and security/safety





Relationship (or trade-off) between privacy and security/safety



Patients vs. Participants

Private industry is increasingly asking hospital systems to consider whether they are willing to share or sell (i.e., ‘commercialize’) biospecimens and health data in both identified forms and de-identified forms (Spector-Bagdady, 2021)



WHAT DOES IT MEAN TO HAVE NO GENETIC PRIVACY?

Academic paper based on Uyghur genetic data retracted over ethical concerns

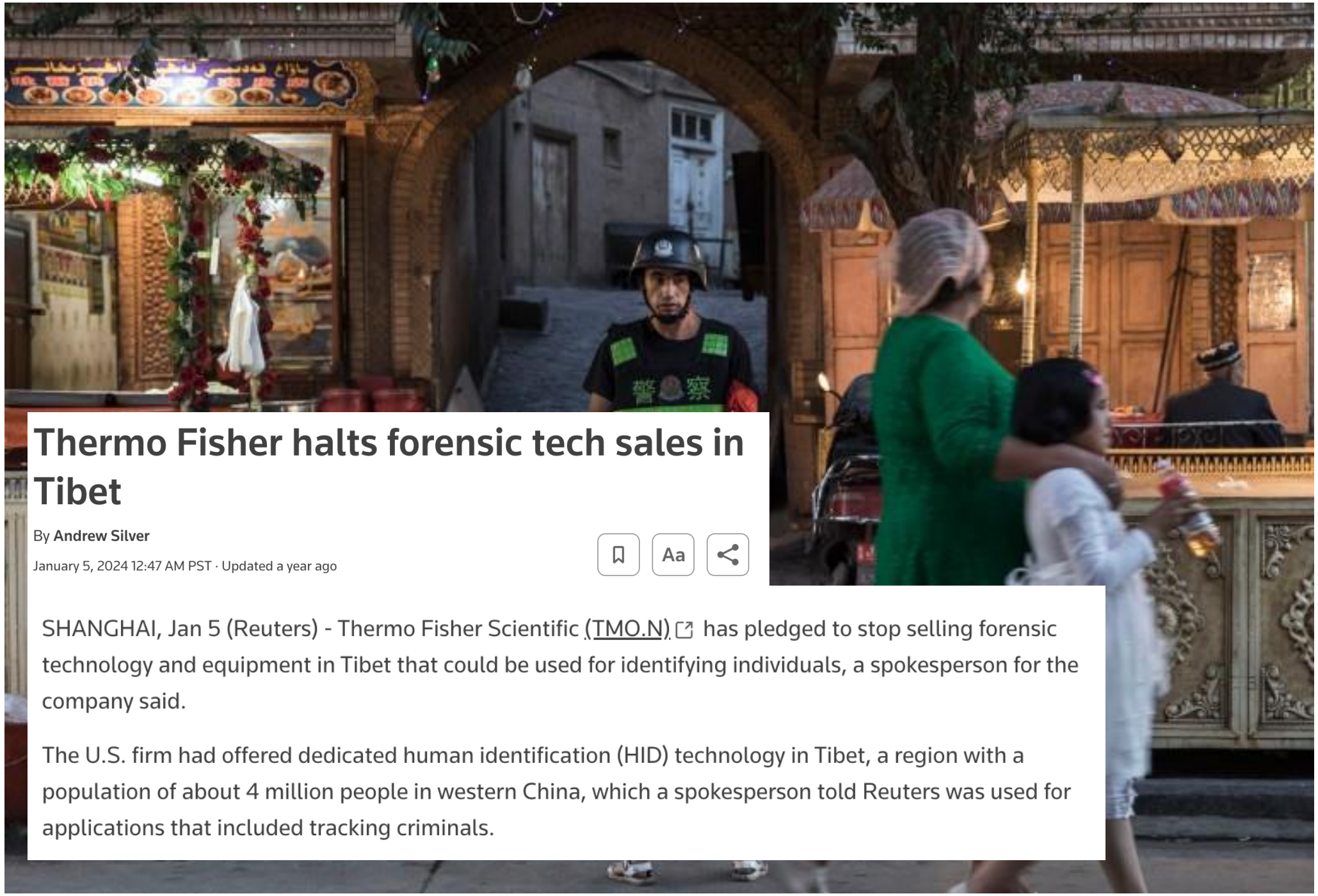
Exclusive: Study published in 2019 used blood and saliva samples from 203 Uyghur and Kazakh people living in Xinjiang capital



Uyghurs are a Turkic ethnic group who are mainly found in Xinjiang. Photograph: Pedro Pardo/AFP/Getty Images



retraction = one of the ten rules of socially responsible science



Thermo Fisher halts forensic tech sales in Tibet

By **Andrew Silver**

January 5, 2024 12:47 AM PST · Updated a year ago

SHANGHAI, Jan 5 (Reuters) - Thermo Fisher Scientific (TMO.N) has pledged to stop selling forensic technology and equipment in Tibet that could be used for identifying individuals, a spokesperson for the company said.

The U.S. firm had offered dedicated human identification (HID) technology in Tibet, a region with a population of about 4 million people in western China, which a spokesperson told Reuters was used for applications that included tracking criminals.

Thailand's new weapon to beat deep-south insurgency - DNA swabs

By **Amy Sawitta Lefevre**

May 6, 2015 3:35 PM PDT · Updated 10 years ago

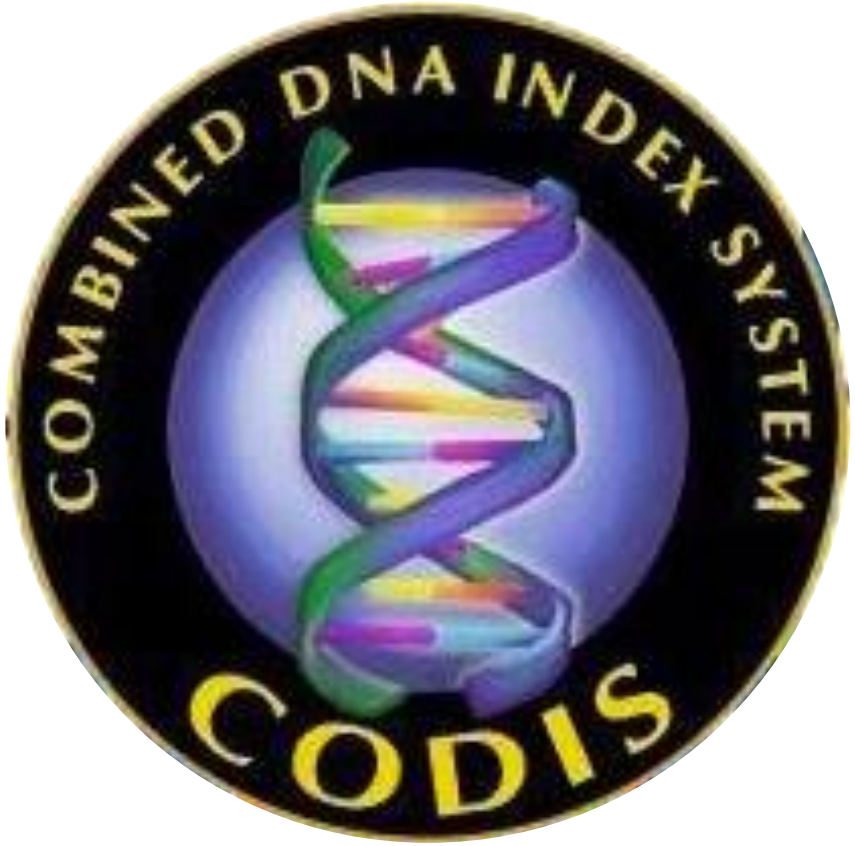
Aa

- The government has been collecting DNA samples to match with biometric data from residents of the southern border provinces
- DNA collection has become a part of the military conscription process for men residing in the southern provinces of Thailand
- Products from Thermo Fisher Scientific, QIAGEN and Promega appear to be used by the Thai police force.



What is "CODIS"?

- CODIS=Combined DNA Index System
- FBI databases, the software used to maintain them and perform analyses, and the genetic loci sequenced for these purposes
- **DNA Identification Act of 1994** approved the creation of an "index," or database that can be searched for genetic matches
    - Used for criminal proceedings and identifying human remains



(Image: Biometric Analysis: Combined DNA Index System (CODIS). *FBI.gov*)

Databases are searched for matches to crime scene DNA

- There are national, state, and local databases (Office of the Inspector General. 2006. *CODIS Audit.*)
- Partial-matches to crime scene DNA are used for familial searches in a number of states (Grabell, Shomron. 2021. *Methods Mol Bio*.)
- After 2013 Supreme Court decision *Maryland v King*, **arrestees**, regardless of conviction status, can have DNA collected and added to CODIS databases in 30 states (*FBI.gov)*



California allows arrestees to be added to CODIS database and, being the most populous state, has the most information in the FBI database

	Total
Arrestee (Not convicted)	909,059
Offender (Convicted)	2,152,679
Investigations aided	102,341

(Data reformatted from : CODIS and NDIS statistics. 2021. *FBI.gov*.)



Relationship (or trade-off) between individual and the collective

Genomic data affects families and populations, not only single persons.

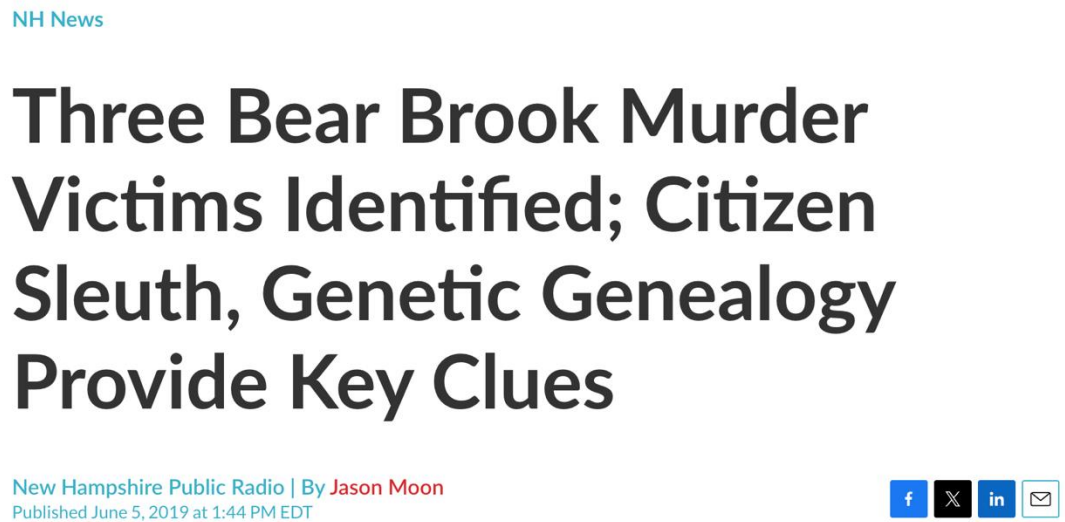


The New York Times

Search for 'Golden State Killer' Leads to Arrest of Ex-Cop

LAPD makes arrest in Grim Sleeper serial killings [Updated]

July 7, 2010 | 12:29 pm

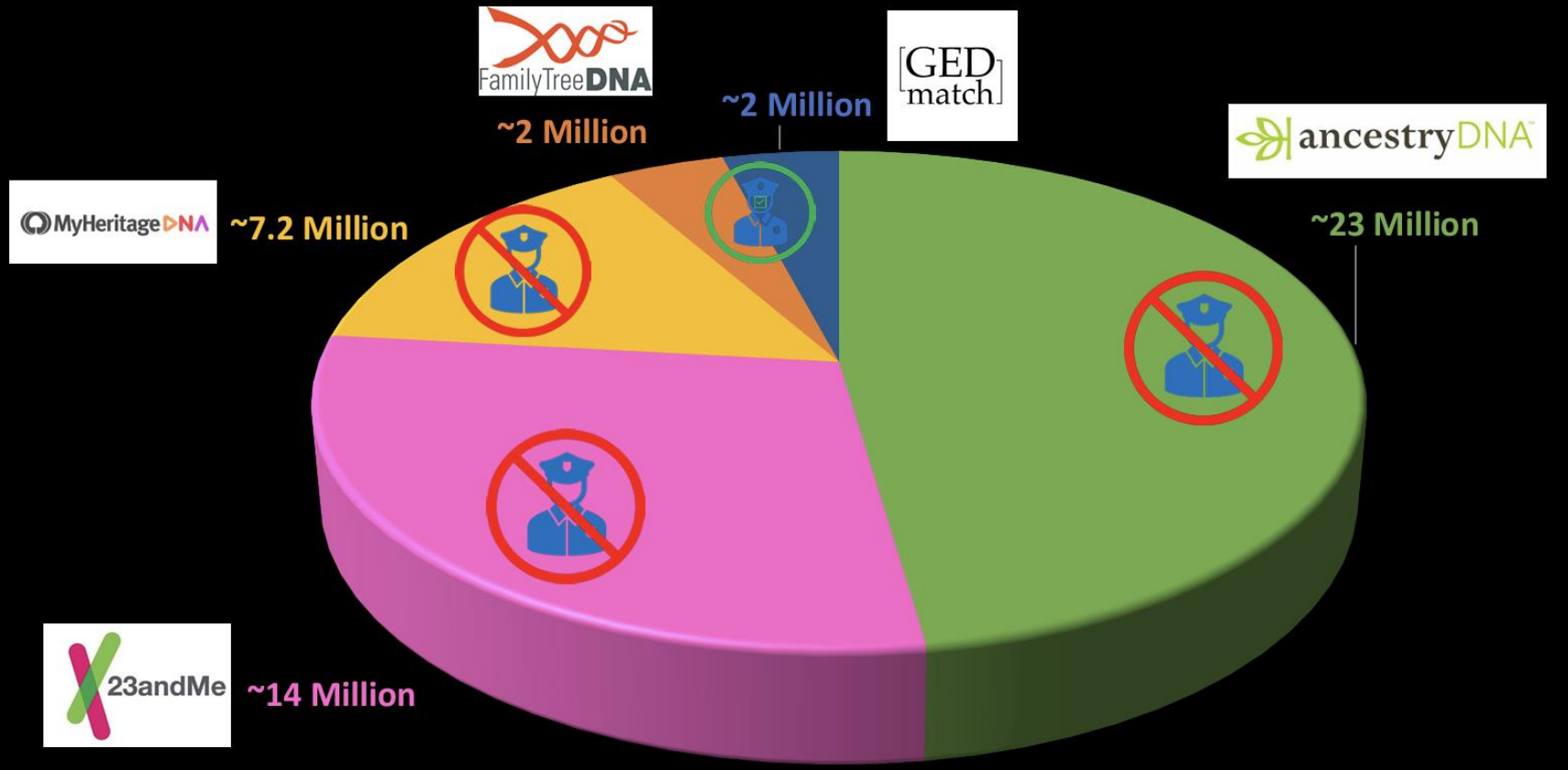


NH News

Three Bear Brook Murder Victims Identified; Citizen Sleuth, Genetic Genealogy Provide Key Clues

New Hampshire Public Radio | By Jason Moon

Published June 5, 2019 at 1:44 PM EDT



Consumer DNA: >48 MILLION PEOPLE

FIGG Databases: <4 MILLION PEOPLE

FamilyTreeDNA

GEDmatch

All users can :

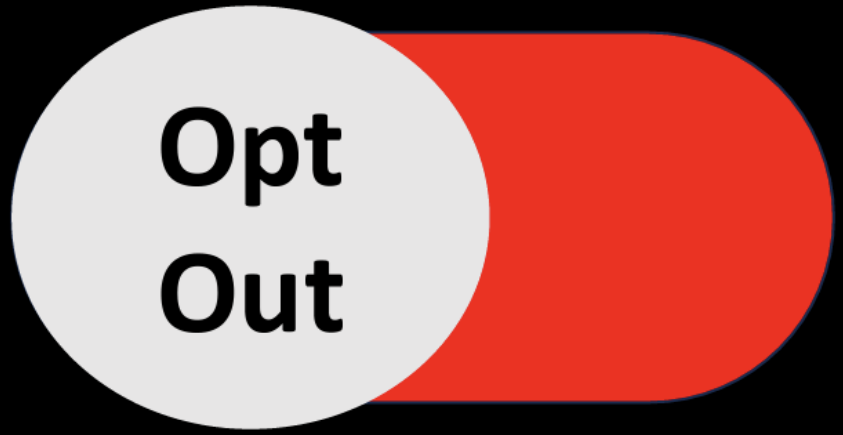


for Law Enforcement Comparisons

FamilyTreeDNA

GEDmatch

All users can :



for Law Enforcement Comparisons

The New York Times

To Identify Suspect in Idaho Killings, F.B.I. Used Restricted Consumer DNA Data

New records show that the F.B.I. identified Bryan Kohberger as a potential murder suspect after tapping consumer databases that were supposed to be off limits.

Listen to this article · 8:28 min Learn more

Share full article 82



Four University of Idaho students were found dead in an off-campus home on Nov. 13, 2022, the victims of a vicious stabbing spree. Margaret Albaugh for The New York Times



By Mike Baker

Feb. 25, 2025

As investigators struggled for weeks to find who might have committed the brutal stabbings of four University of Idaho students in the fall of 2022, they were focused on a key piece of

What do you consider most important?

Autonomy

Privacy

Safety/Security

Individualism

Collectivism

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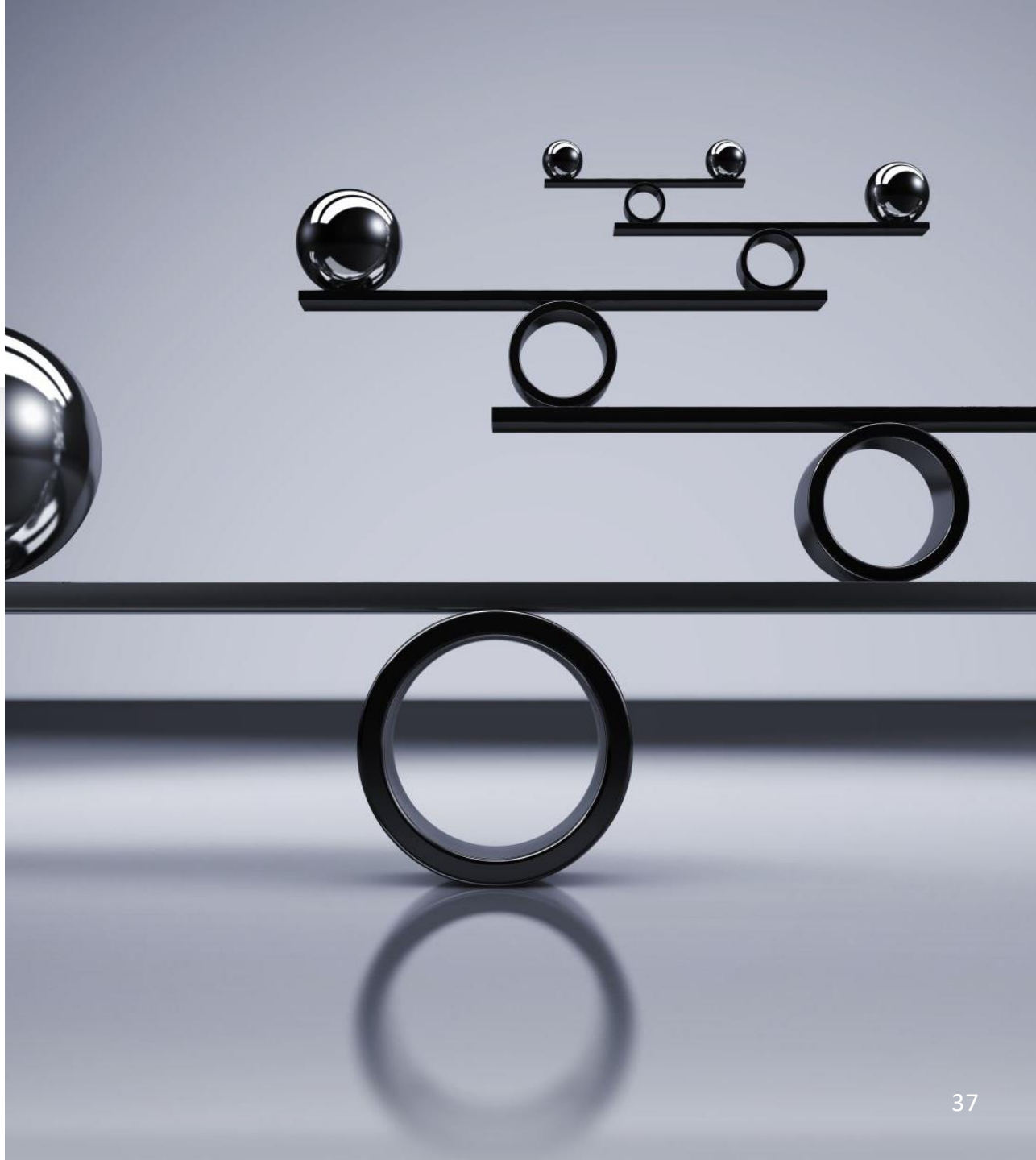
Broader data collection =/= justice

Bias

- **In AI:** An unfair systematic error that leads to an incorrect estimate of effect or association

- **Broadly:** prejudice in favor of or against one thing, person, or group compared with another, usually in a way considered to be unfair

Discrimination is underlying ethical concern related to bias



“Considering the **disproportionate representation of racial minorities in law enforcement DNA databases**, certain individuals have a higher risk of being targeted for investigation simply because genetic relatives are represented in the CODIS database.” (federaljudicialcenter.gov)

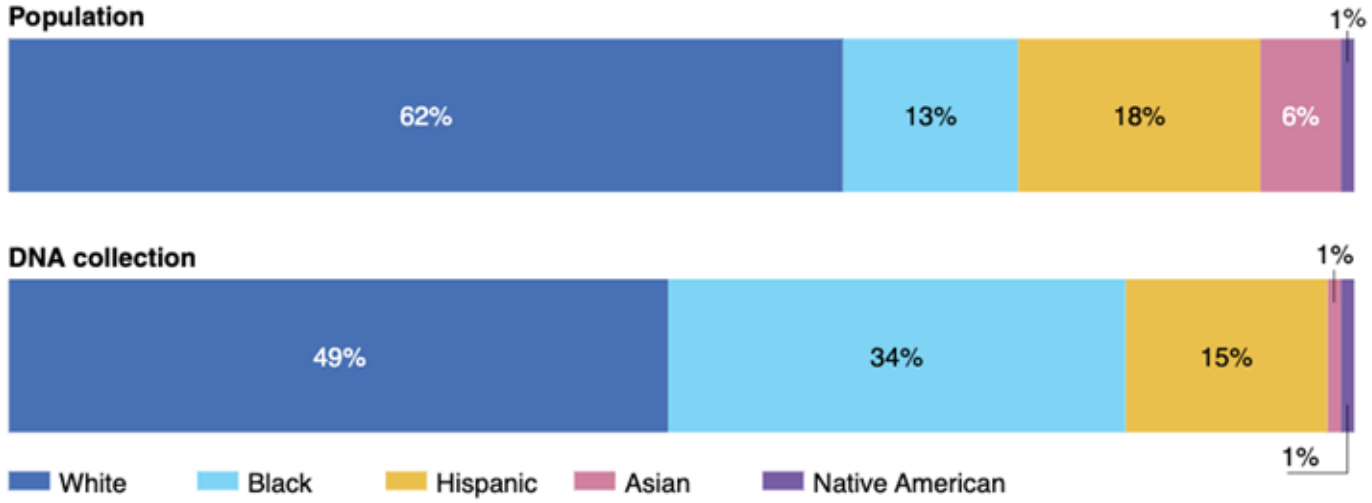


Fig. 1 | Comparison of US population with DNA collection for NDIS deposition, categorized as defined by the US Census. Data from ref. [3].

(Images: Court, Denise. 2021. *Nature Computational Science*.)

A BIASED PICTURE?

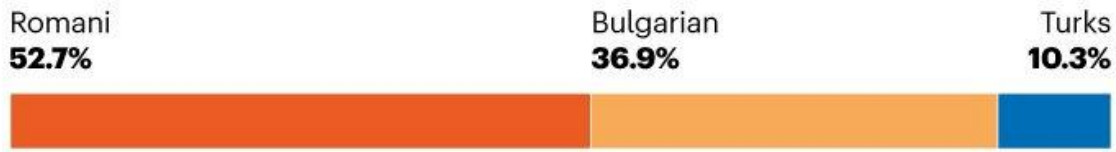
Roma people are over-represented in databases such as the YHRD*, which is accessed by law-enforcement agencies. Without thorough evaluation, the effects of this over-representation are unclear.

Bulgaria

Population (2011 census†)



Samples in YHRD



Hungary

Population (2016 microcensus‡)



Samples in YHRD



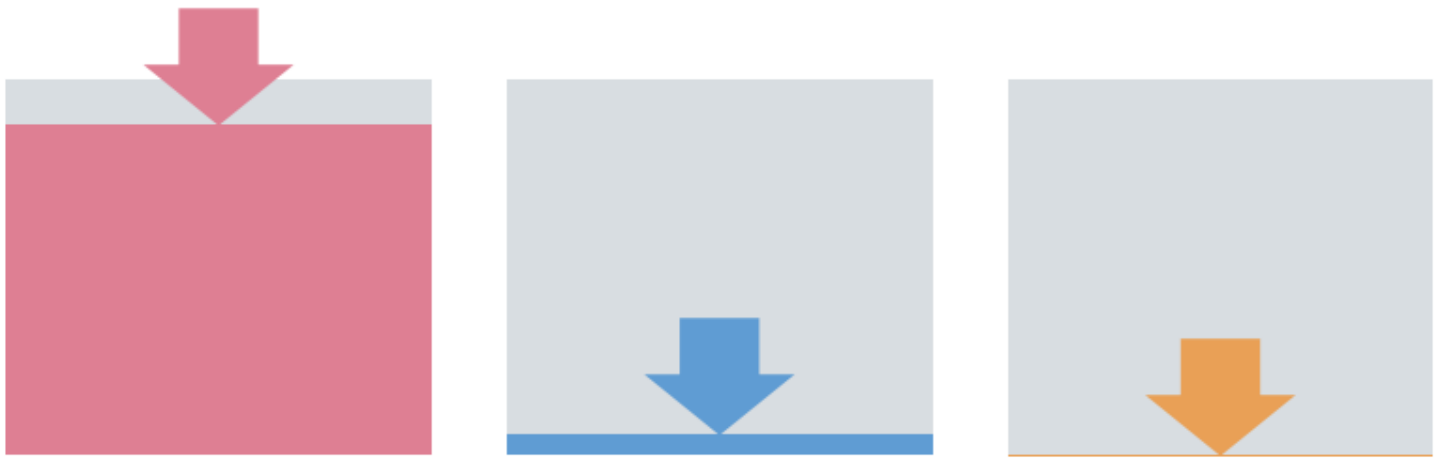
*YHRD, Y-STR Haplotype Reference Database. **All ethnic-group labels are those used by the YHRD and population censuses. YHRD data as of 11 November 2021. †Total is <100%: some respondents did not identify with an ethnic group. ‡Total is >100%: respondents could select more than one ethnic category.

©nature

GWAS
Diversity Monitor

Total GWAS participants diversity

Version 1.0.0. Last check for data: 2025-09-05 00:34:26 .



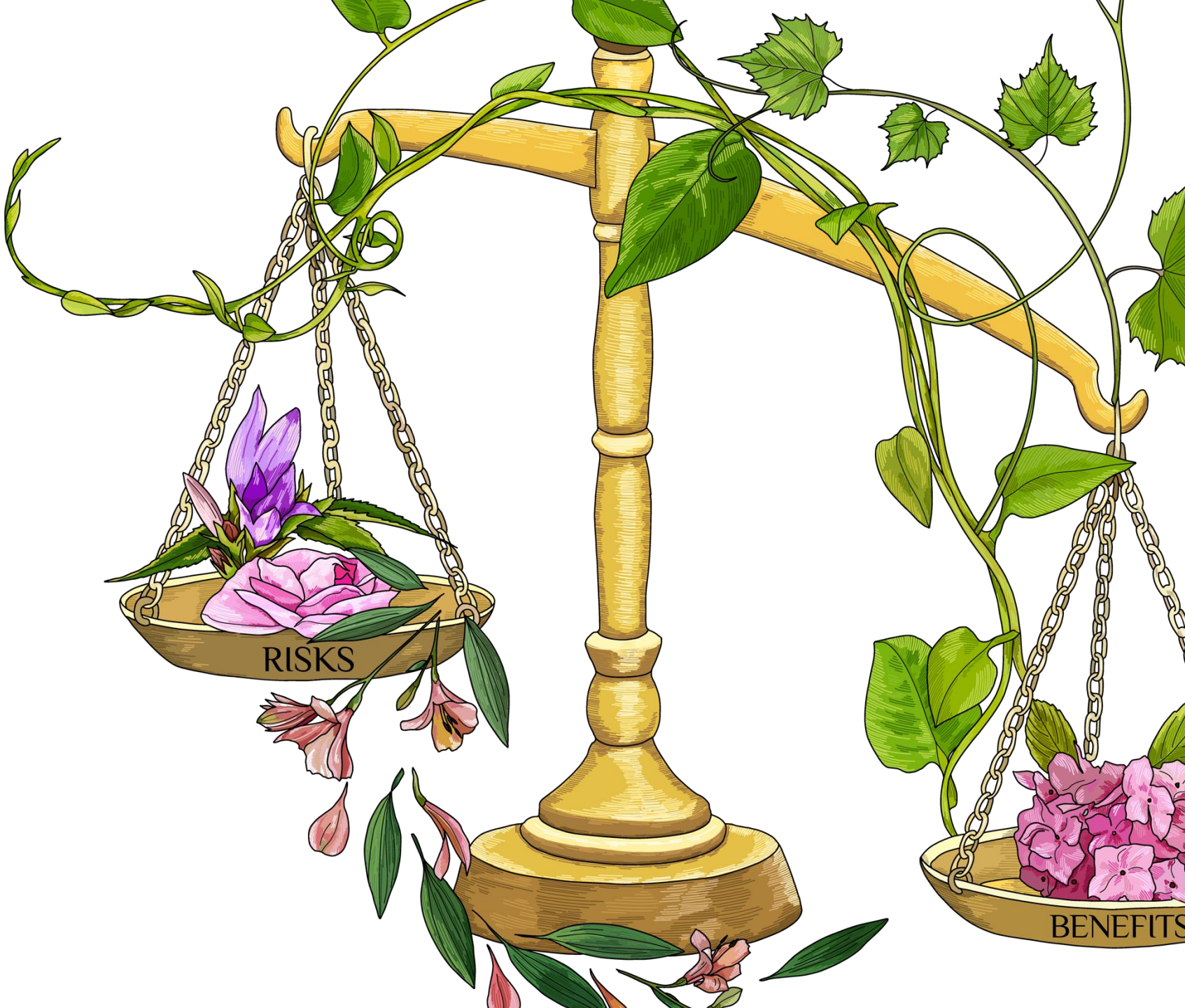
87.77%	5.33%	0.16%
European	Asian	African

fair genomic surveillance requires:

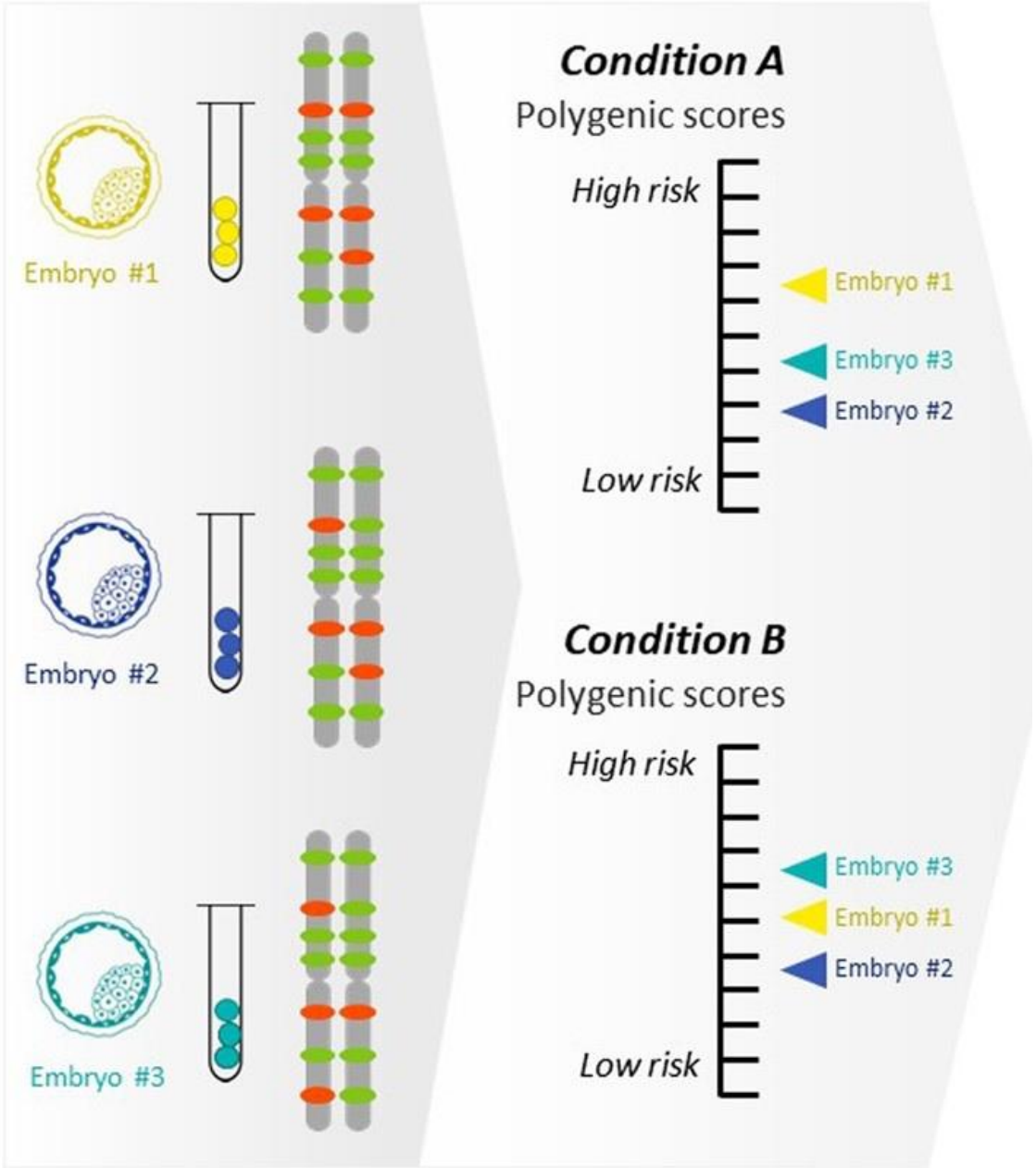
- transparent governance
- community participation
- limits on secondary use
- equitable benefit sharing
- anti-discrimination protections
- accountability mechanisms across borders

Genomic data as a public good

Genomic data as a private good



Polygenic embryo screening



Hum Reprod Update, Volume 30, Issue 5, September-October 2024, Pages 529–557, https://doi.org/10.1093/humupd/dmae012

A growing industry



Genomic Prediction

2017



2020

Orchid Health

2019



Nucleus Genomics

2021

*2017**

MyOme







2023

Herasight*

* Heliospect is the research arm

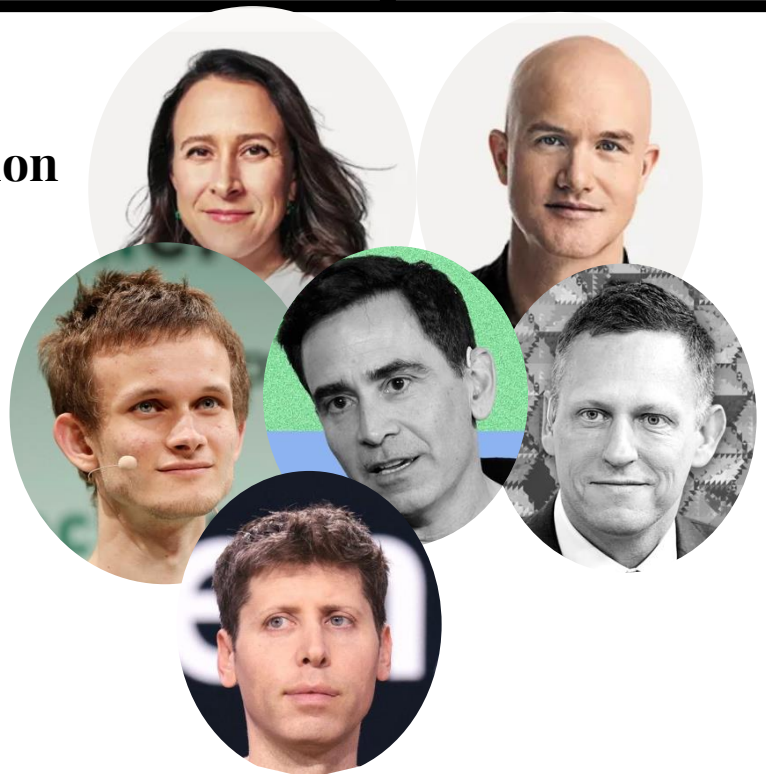
* Didn't initially offer PGT-P

Significant investment from tech titans



2017

Genomic Prediction



Orchid Health

2019





2021

Nucleus Genomics

Herasight

2023





pickyourbaby.com

Nucleus IVF+

Have a smarter baby.



THESE BABIES HAVE GREAT GENES

Nucleus IVF+

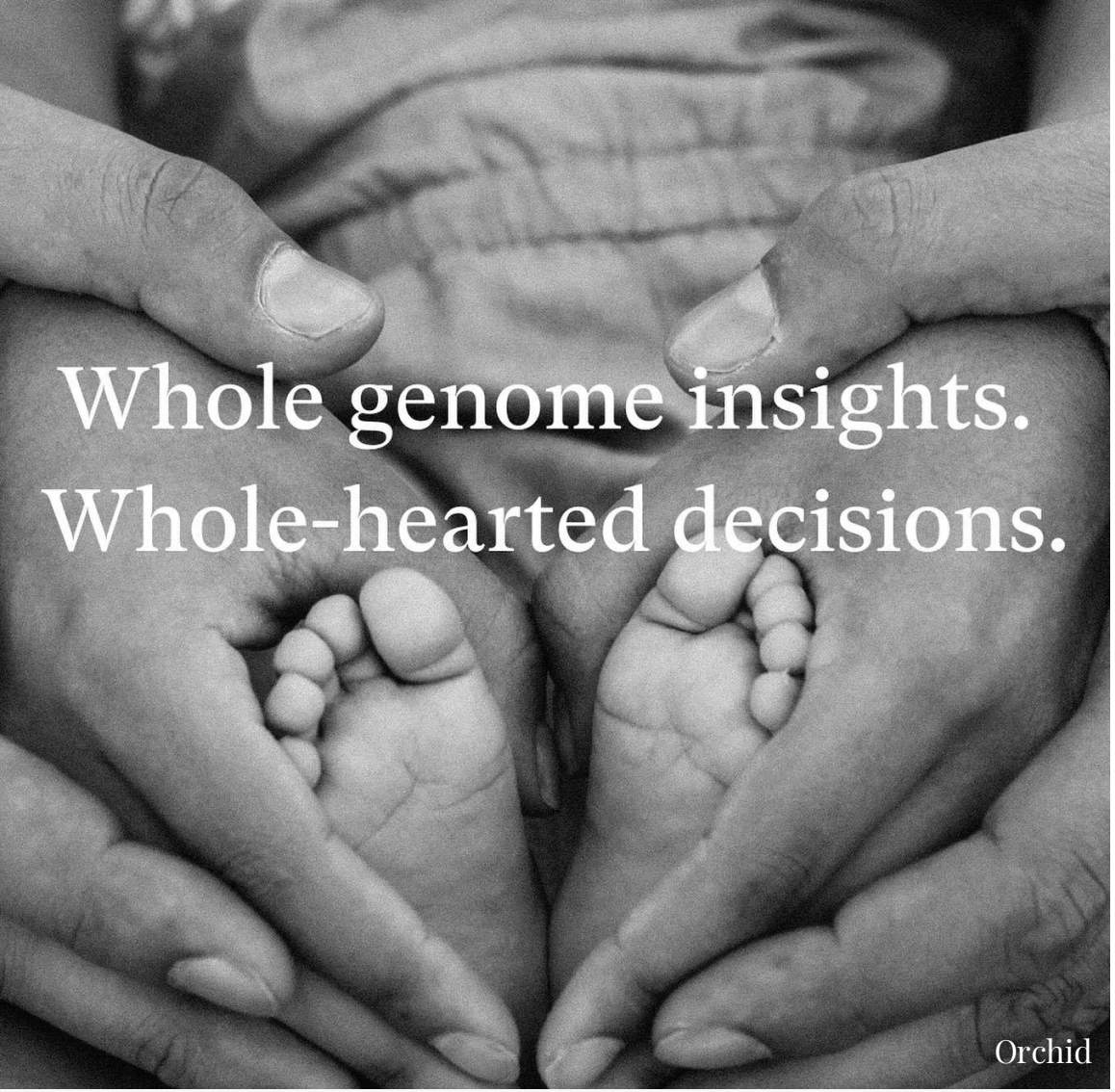


pickyourbaby.com

Nucleus IVF+

Have a taller baby.





Parental autonomy & informed choice

choice. better than chance.

Genomic Prediction

Make an informed decision

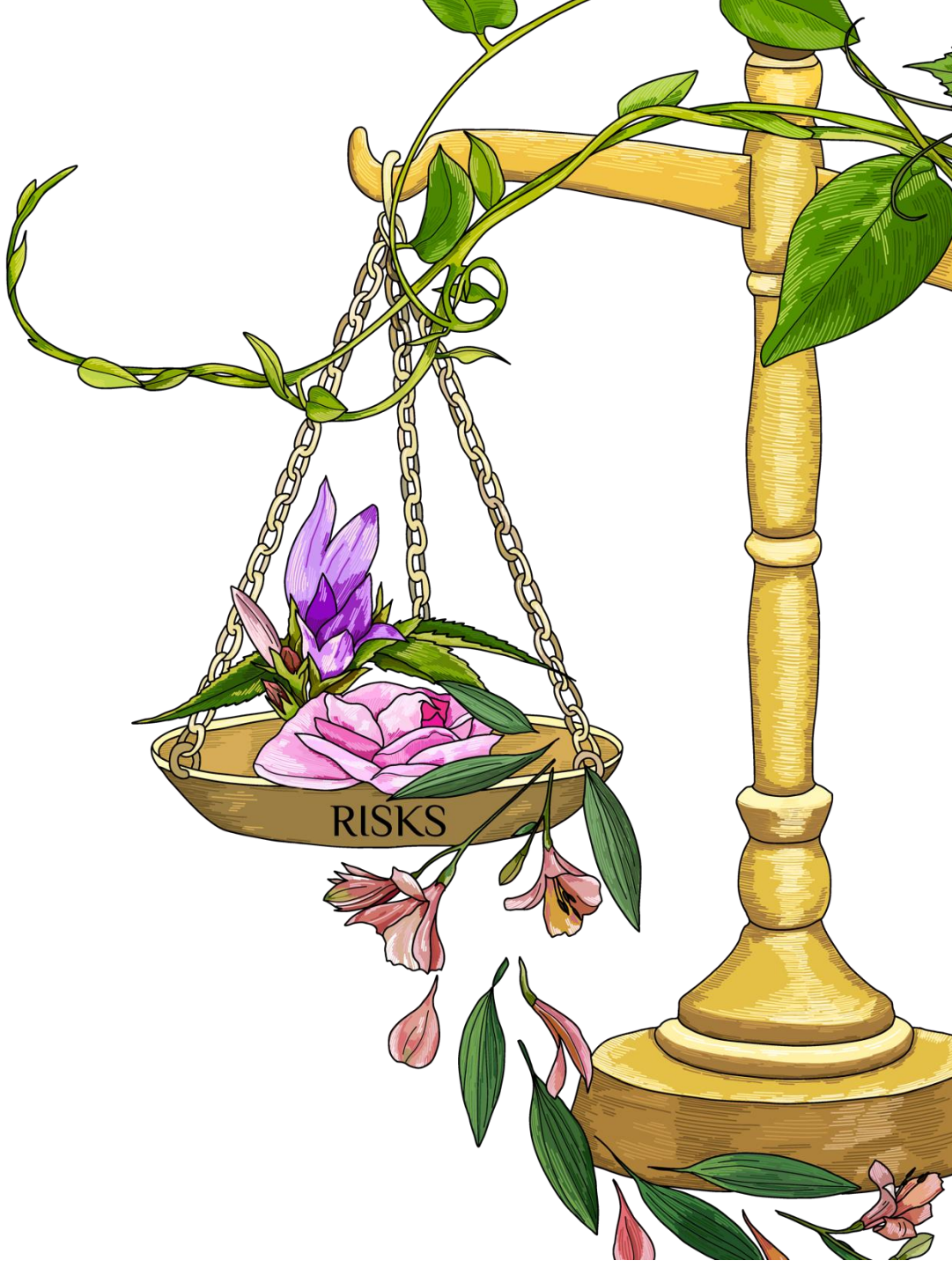
Orchid

We understand that genetics is complex and deeply personal. That's why genetic counseling is central to our approach—we don't just present information, we help parents find meaning in the context of their unique situation and values. We want every individual to feel informed and confident when working with us.

Herasight

Leaving you with a question

How do justice concerns shift when genomic data is considered a public good versus a private good?



Faden & Powers

1. Would people without genomic surveillance be so deprived that they could not live decent lives?

2. Does not providing access to genomic surveillance technologies lead to the kinds of disadvantages that are so systematically oppressive such that people cannot have decent lives?

